

Bus Reservation – Seat Booking System

Feature Set I

1. Requirements

1.1 Introduction

The Bus Reservation – Seat Booking System is a system used to manage bus seat reservations. It stores passenger details, bus details, route information, seat availability, and booking details. It helps users select an available seat and generate booking confirmation.

1.2 Objective of the System

The main objectives are:

To store passenger information.

To maintain bus and route details.

To display available seats.

To allow passengers to select seats.

To record and manage bookings.

To generate booking confirmation.

To reduce manual work and booking errors.

1.3 Functional Requirements

The system should provide the following functions:

Passenger Details – Enter and store Passenger ID, name, age, gender, and phone number.

Bus Details – Store bus ID, bus name, bus number, and total seats.

Route Details – Store source, destination, departure time, and arrival time.

Seat Selection – Display available seats and allow the passenger to select a seat.

Seat Availability – Check whether the selected seat is available.

Booking – Store passenger, bus, route, and seat information for the booking.

Booking Confirmation – Confirm the booking and generate booking details.

1.4 Non-Functional Requirements

Easy to use: The system should have a simple interface.

Fast: It should process bookings quickly.

Reliable: Booking information should be stored correctly.

Secure: Passenger and booking data should be protected.

Accurate: The system should avoid duplicate seat bookings.

Maintainable: The system should be easy to update.

1.5 Hardware Requirements

Computer or laptop

Minimum 4 GB RAM

Minimum 20 GB available storage

Keyboard and mouse

Monitor

Internet connection, if required by the application

1.6 Software Requirements

Operating System: Windows/Linux

Programming language: C / C++ / Java / Python, as selected for the project

Database: MySQL or another suitable database

IDE or code editor

Database management software, if required

1.7 User Requirements

The user should be able to:

Enter passenger details.

View bus and route information.

Check available seats.

Select a seat.

Enter booking details.

Confirm the booking.

View booking confirmation.

1.8 System Constraints

The system requires valid passenger and booking information.

A seat cannot be booked if it is already reserved.

Each passenger should provide the required details.

The system depends on correct bus and route data.

The database must be available to store booking information.

2. Algorithm

Algorithm: Bus Reservation – Seat Booking System

Start

Enter passenger details such as Passenger ID, name, age, gender, and phone number.

Enter bus details such as Bus ID, bus name, bus number, and total seats.

Enter route details such as source, destination, departure time, and arrival time.

Display the available seats.

Select the required seat.

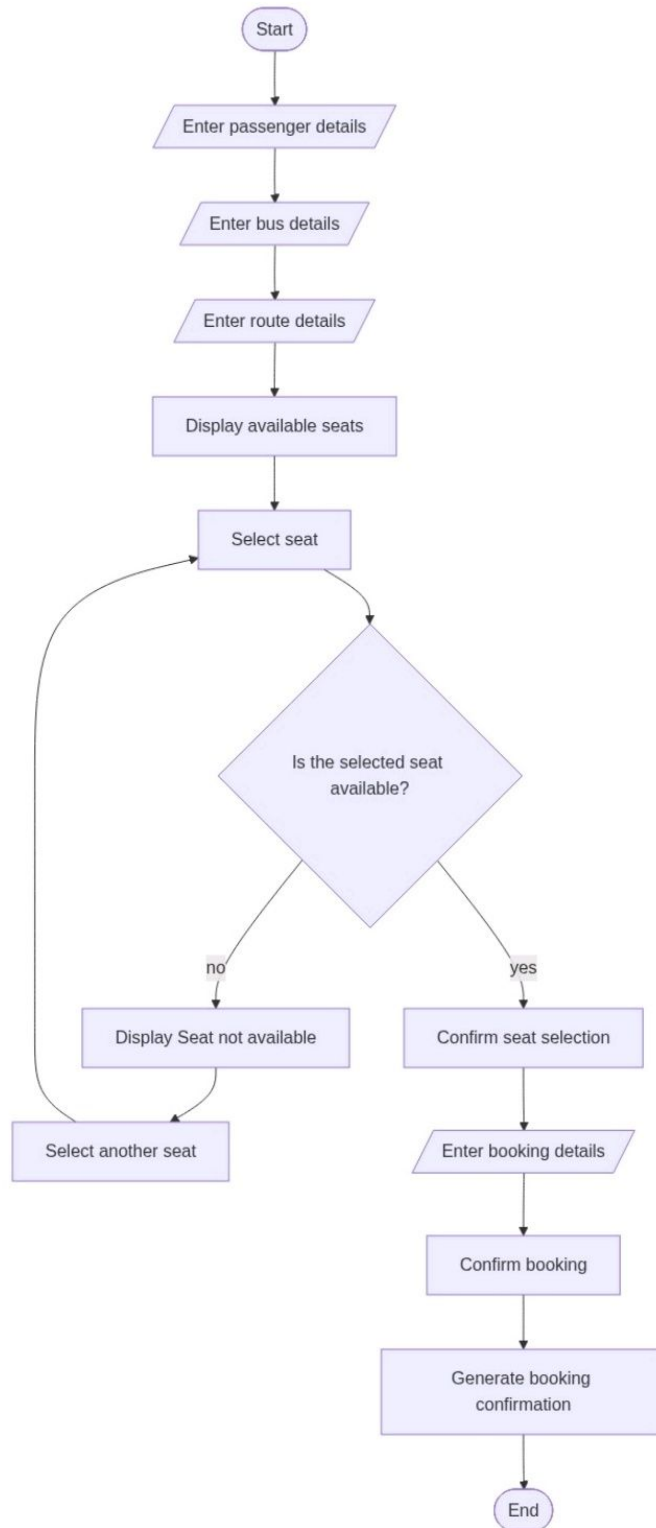
Enter booking details.

Confirm the booking.

Generate and display the booking confirmation.

End

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